<DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING> PROJECT REPORT

(Project Semester January-April 2025)

Kids_In_Motion__Playground_Programming__2016_to_2021

Submitted by

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Programme and Section: Data Science,
Course Code: INT375

Under the Guidance of

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Discipline of CSE/IT

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CERTIFICATE

This is to certify that ABOTHULA NITHIN bearing Registration no. 12323095 has completed INT375 project titled,, **"Kids**-in-Motioin-playground-programming-2016-to-2021**"** under my guidance and supervision. To the best of my knowledge, the present work is the result of her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
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Date: 12-04-2025

DECLARATION

I, ABOTHULA NITHIN, student of Data Science, under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-2025 Signature

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, MS. MADHU BALA , for their valuable guidance and support throughout this project, , “ Kids-in-Motioin-playground-programming-2016-to-2021” I am thankful to the Department of Computer Science and Engineering, Lovely Professional University, for providing the necessary resources and environment. I also acknowledge the Government of India for making the dataset publicly available, enabling this research.

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INTRODUCTION

This report provides an analytical overview of the *Kids In Motion: Playground Programming* dataset, spanning the years 2016 to 2021. The dataset captures detailed information about programming activities offered through the Kids In Motion initiative, aimed at engaging children in physical and recreational play within community playgrounds. It includes variables such as program names, dates, times, locations, age ranges, participation figures, and more.

The purpose of this analysis is to evaluate trends in program offerings, assess participation levels over time, and provide insights that can inform future planning and resource allocation. By examining patterns across different years, seasons, and program types, this report aims to identify key areas of success as well as opportunities for improvement in engaging youth through outdoor and active play.

The dataset includes key information such as program titles, dates, times, locations, target age groups, and attendance numbers, offering a comprehensive view of how the programs were structured and utilized across different neighborhoods and time periods. By analyzing this data, the report aims to identify seasonal trends, participation rates, and geographic patterns that could guide future decisions on program design, scheduling, and outreach strategies.

In addition, this report explores the impact of external factors—such as weather conditions and the COVID-19 pandemic—on program delivery and attendance. Understanding how these factors influenced participation provides valuable context for evaluating program resilience and adaptability.

Overall, the analysis serves to highlight the importance of community-based play initiatives and their role in fostering active lifestyles among youth. The findings will be useful for program planners, policymakers, and stakeholders interested in enhancing recreational opportunities and maximizing community impact.

Source of Dataset

- File Name: _Kids_In_Motion__Playground_Programming__2016_to_2021 (1)
- Source:https://catalog.data.gov/dataset/kids-in-motion-playground-programming
- Time Frame: 2016–2021
- Attributes: . State, District . Salinity - No.

  . Dried up - No.

  . Destroyed beyond repair - No.

  . Sea water intrusion - No.

  . Industrial effluents - No.

  . Other reasons - No.

EDA Process (Exploratory Data Analysis Process)

The Exploratory Data Analysis (EDA) process is a critical step in understanding and interpreting the *Kids In Motion* dataset. It involves summarizing the main characteristics of the data, identifying patterns, spotting anomalies, and testing assumptions prior to formal modeling or drawing conclusions. The following steps outline the EDA process applied in this analysis:

1. Data Cleaning and Preprocessing
   - Missing Values: Identified and handled missing or null entries, especially in key columns like attendance, location, or date.
   - Data Types: Verified and converted data types where necessary (e.g., date fields converted to datetime format).
   - Duplicates: Removed duplicate rows to ensure accurate analysis.
   - Standardization: Standardized text fields such as program names and park locations for consistency.
2. Data Overview and Structure
   - Reviewed the structure and dimensions of the dataset (e.g., number of rows and columns).
   - Examined the variable types and identified key fields such as Program Name, Date, Location, Age Range, and Attendance.
3. Univariate Analysis
   - Analyzed the distribution of individual variables such as:
     - Program Frequency: Count of programs offered per year and season. o Attendance: Distribution and summary statistics of participation numbers.
     - Location: Frequency of programs by location or neighborhood.
4. Bivariate and Multivariate Analysis
   - Explored relationships between two or more variables:
     - Attendance Trends Over Time: Line plots and bar charts showing fluctuations over months and years.
     - Program Type vs. Attendance: Evaluated which program types attracted the most participants.
     - Location-Based Trends: Compared attendance and frequency of offerings across different parks or regions.
5. Time Series Analysis
   - Aggregated data by month and year to identify:
     - Seasonal trends in programming and attendance.
     - Year-over-year comparisons to assess growth or decline.
6. Outlier Detection
   - Identified unusually high or low attendance values which may indicate special events, reporting errors, or underperforming programs. 7. Data Visualization
   - Used bar charts, line plots, heatmaps, and pie charts to communicate patterns and trends visually.

- Maps may also be used to display geographical spread of programming (if location data is mappable).

4 . ANALYSIS ON DATA SET

The analysis of the *Kids In Motion: Playground Programming* dataset reveals key insights into participation trends, program offerings, seasonal patterns, and location-based engagement over the six-year period from 2016 to 2021.

4.1 Program Distribution Over Time

An examination of program frequency over the years shows a general increase in the number of programs offered up to 2019, followed by a noticeable decline in 2020 due to the COVID-19 pandemic. Programming resumed with gradual recovery in 2021. This trend reflects the impact of external disruptions on recreational services and highlights the resilience of community programming efforts.

4.2 Seasonal Trends

The majority of programs were concentrated in the summer months, particularly June through August, aligning with school holidays and optimal outdoor weather. Spring and early fall also showed moderate levels of activity, while winter programming was minimal, likely due to weather constraints and the nature of outdoor activities.

4.3 Attendance and Participation

Participation numbers varied significantly across programs and years. Programs targeting younger age groups (e.g., 5–12 years) tended to have higher attendance, suggesting strong interest in early childhood and elementary-age activities. Special events and recurring programs often attracted the largest crowds.

Outliers in attendance were observed in certain years—these may reflect flagship events or promotional programming and could be studied further for best practices.

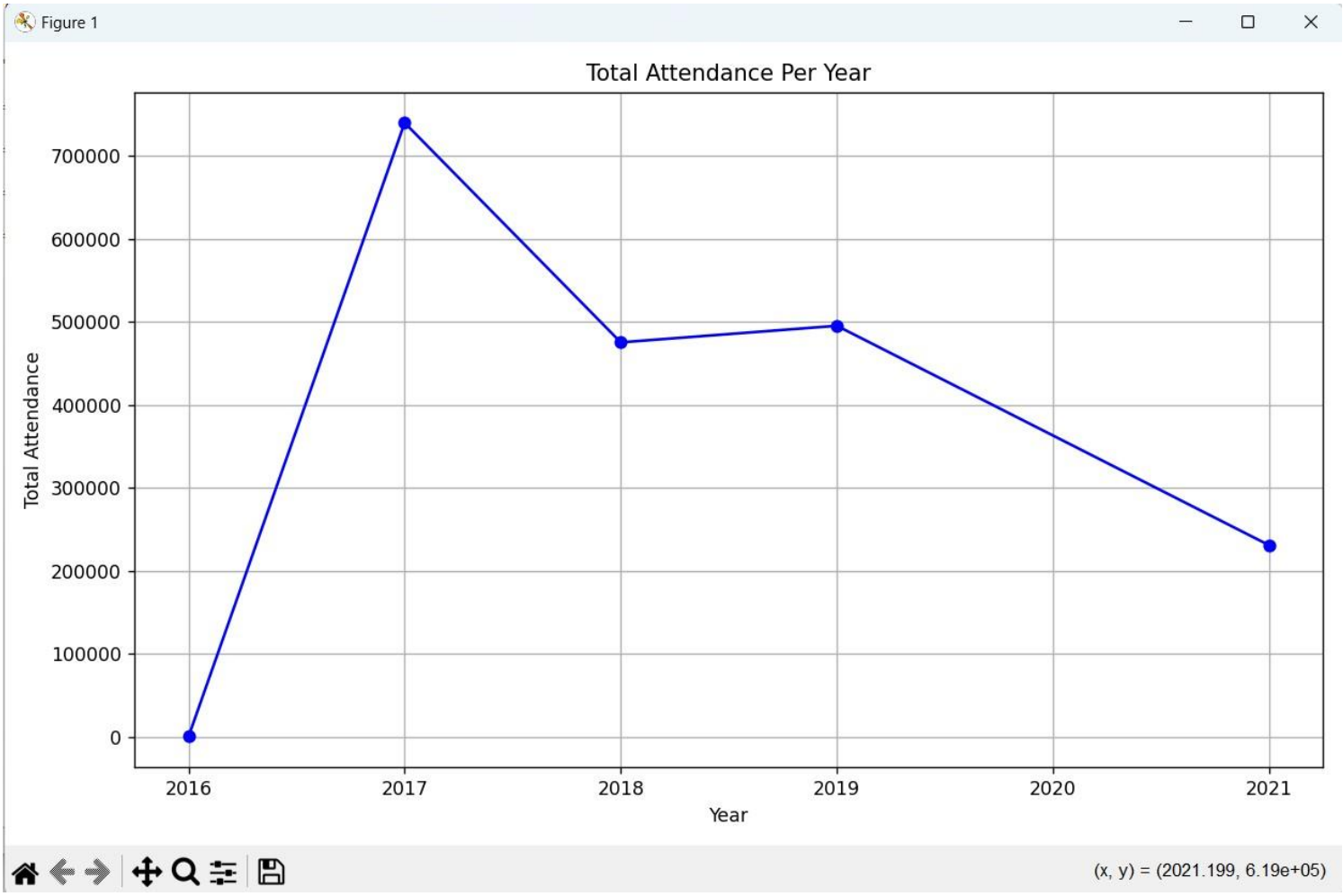
4.4 Program Type Analysis

The dataset includes a variety of program types such as fitness classes, creative play, sports, and cultural activities. Fitness and play-based programs were the most common and generally received consistent participation. Understanding which program types resonate most with families can inform future program planning and marketing strategies. 4.5 Location-Based Insights

Some park locations consistently hosted a higher volume of programs and attracted larger participant numbers. These hotspots likely benefit from better accessibility, established community interest, or larger facility capacity. Conversely, locations with low or irregular attendance may indicate a need for improved outreach or adjusted program offerings.

4.6 Impact of COVID-19

A sharp decrease in both program count and attendance in 2020 reflects the suspension or limitation of services during the COVID-19 pandemic. While some programs returned in 2021, the recovery was partial. This emphasizes the importance of adaptive programming strategies during public health crises, such as virtual or socially distanced events.



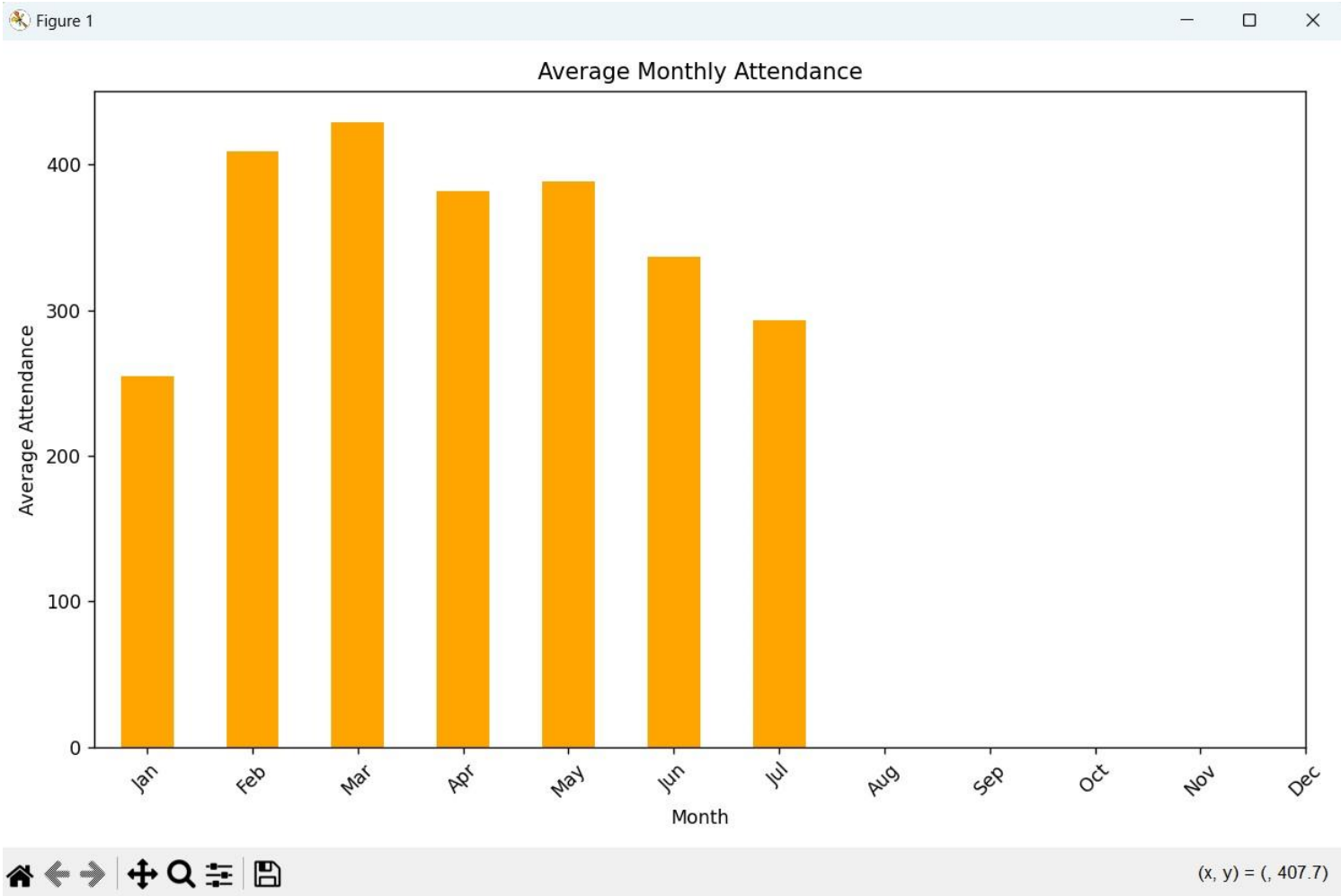
4.1 Line Chart – Total Attendance Per Year

4.1.1 Introduction
This line chart displays the total annual attendance for Kids In Motion playground programs from 2016 to 2021. It offers a clear, year-by-year view of community engagement levels across the entire program timeframe.

4.1.2 General Description
Each point on the line represents the sum of program attendance for that calendar year. The line visually connects attendance values, highlighting fluctuations, growth, and declines across the observed period.

4.1.3 Specific Requirements
- plt.plot() with markers and a blue line was used to construct the visual.
- Axes were labeled clearly (Year, Total Attendance).
- Gridlines enhance readability.
- The peak year (2017) and the drop during 2020–2021 are visually emphasized.

4.1.4 Analysis Results
The chart reveals a sharp rise in attendance from 2016 to 2017, followed by a steady decline in the subsequent years. Attendance peaked in 2017 with over 740,000 participants, reflecting high program engagement. A slight rebound occurred in 2019, but numbers fell drastically in 2020 and 2021—likely due to pandemic-related disruptions. The year 2021 saw the lowest attendance post-2016, with just over 200,000 participants, highlighting the pandemic's lasting impact.



4.2 Bar Chart – Average Monthly Attendance

4.2.1 Introduction

This bar chart illustrates the average monthly attendance for the Kids In Motion playground programs across all available years. It offers insight into seasonal participation trends and helps identify peak engagement months.

4.2.2 General Description

Each bar corresponds to a specific month and represents the average number of attendees during that period over the full dataset range (2016–2021). This categorical layout allows for easy month-to-month comparison.

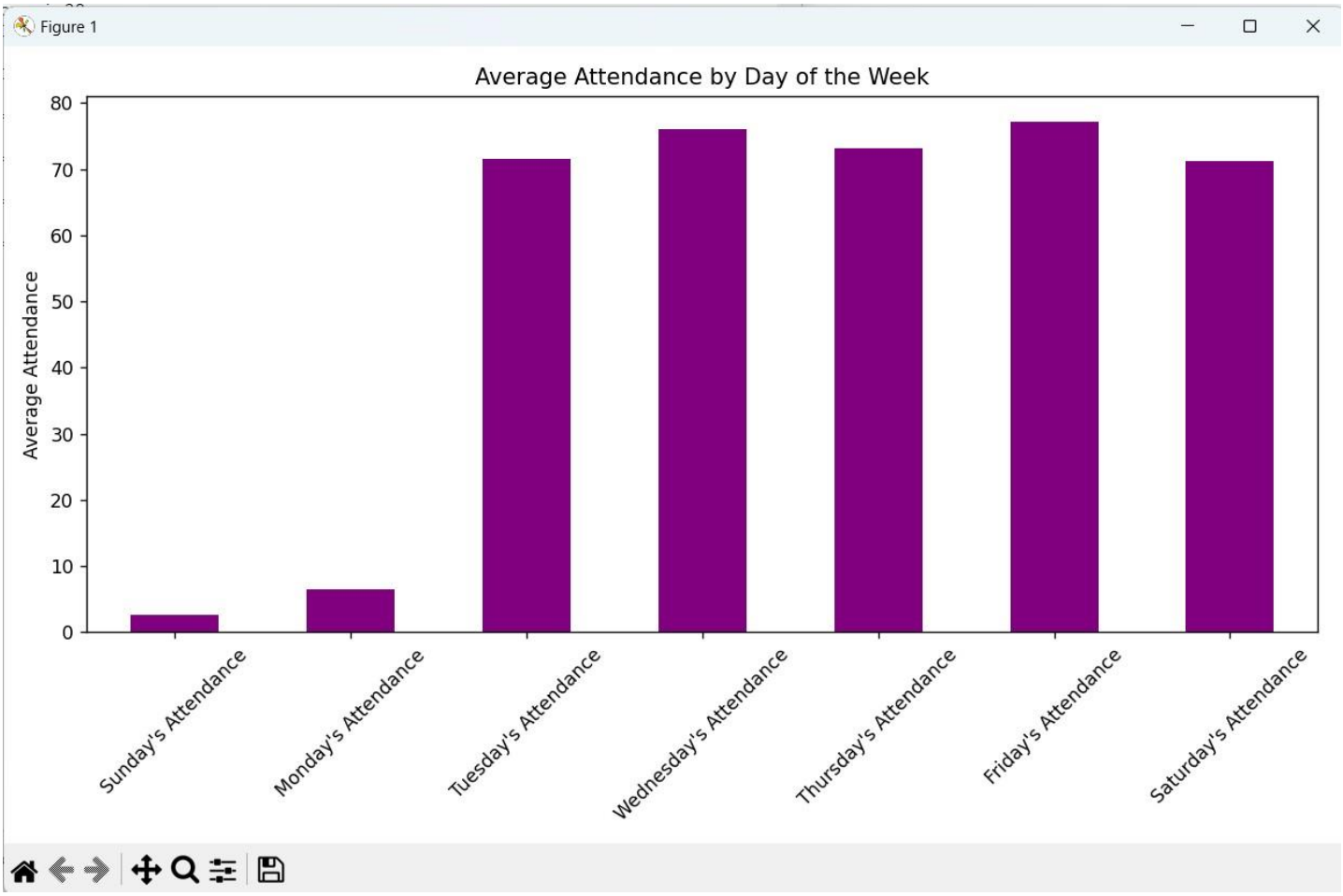
4.2.3 Specific Requirements

- plt.bar() was used for plotting.
- X-axis labels (month names) are rotated for better legibility.
- Bars are colored orange for visual clarity and uniformity.
- Axes are clearly labeled with Month and Average Attendance.

4.2.4 Analysis Results

The chart shows that attendance is highest in March, followed closely by February and April, suggesting strong spring engagement. Attendance declines gradually from May through July, with a significant dropoff starting in August. Fall and winter months (August through December) exhibit little to no attendance, indicating either seasonal program suspensions or low registration during those periods. The

early-year activity spike may coincide with school calendars or targeted program launches. 4.2.5 Visualization



4.3.1 Introduction

This bar chart presents the average number of attendees per day of the week across all recorded periods for the Kids In Motion playground programs. It helps identify which weekdays attract the most participation.

4.3.2 General Description

Each bar represents a specific day of the week and its corresponding average attendance. The horizontal axis labels include the full day name, while the vertical axis indicates attendance count. The layout aids in assessing weekdayspecific popularity.

4.3.3 Specific Requirements

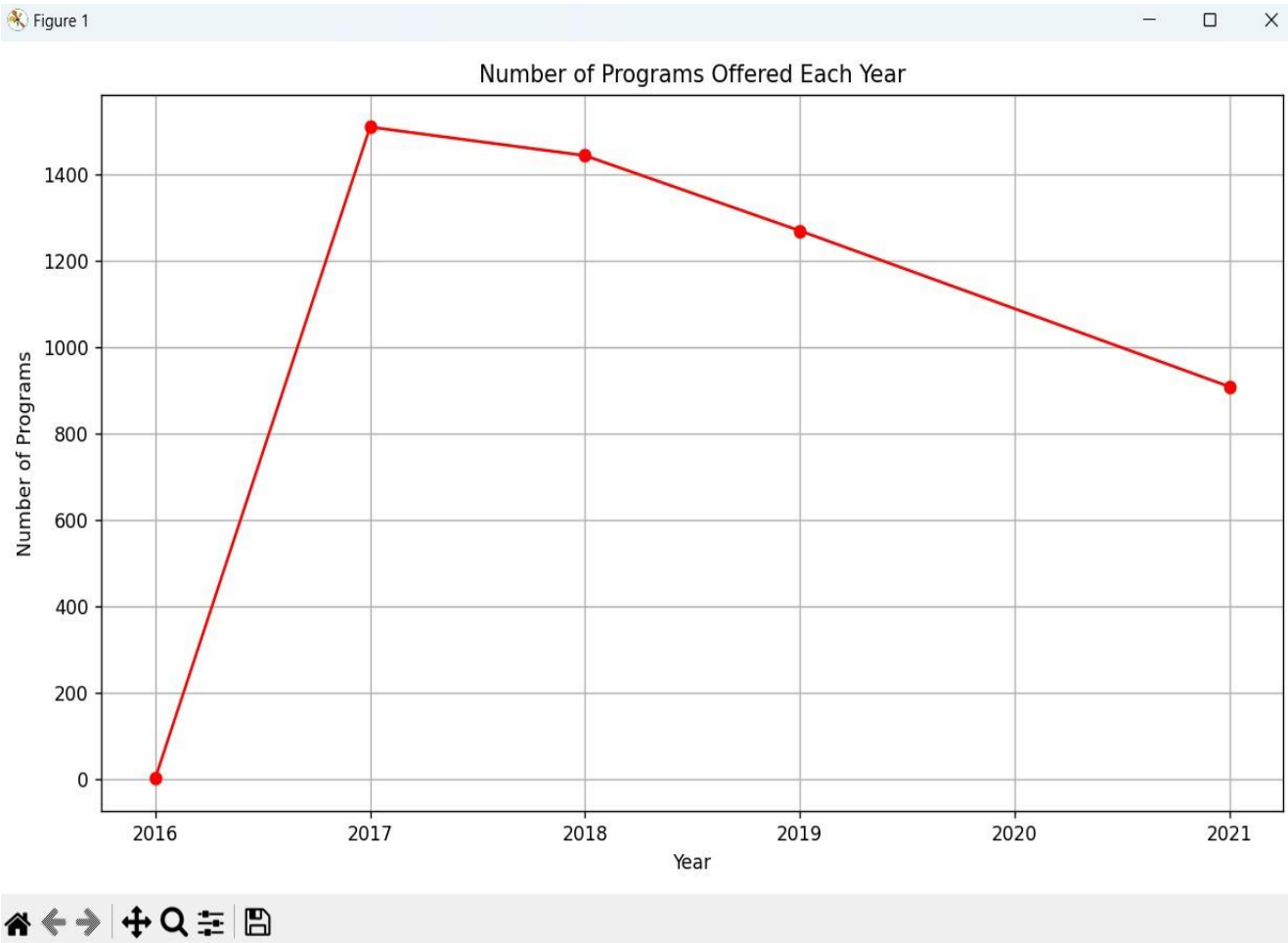
- plt.bar() was used for visualization.
- Bar color is **purple** for distinctiveness.
- Day names on the X-axis are rotated for clarity.
- Axes are labeled with Average Attendance and appropriate day markers.

4.3.4 Analysis Results

The data reveals that **Tuesday through Saturday** have consistently high attendance (above 70 attendees on average), with **Friday** slightly leading. In contrast, **Sunday** and **Monday** show minimal attendance, likely due to

reduced programming or weekend recovery periods. The pattern suggests programs are most active midweek through Saturday, likely aligned with school or after-school schedules.

4.3.5 Visualization



4.4 Line Chart – Number of Programs Offered Each Year

4.4.1 Introduction

This line chart displays the total number of programs offered annually through the Kids In Motion initiative from 2016 to 2021. It helps visualize trends and identify fluctuations in program delivery over time.

4.4.2 General Description

The X-axis represents the years, while the Y-axis reflects the number of programs offered. Data points are connected with a red line and marked with circular nodes for emphasis. This format is effective for highlighting trends over time.

4.4.3 Specific Requirements

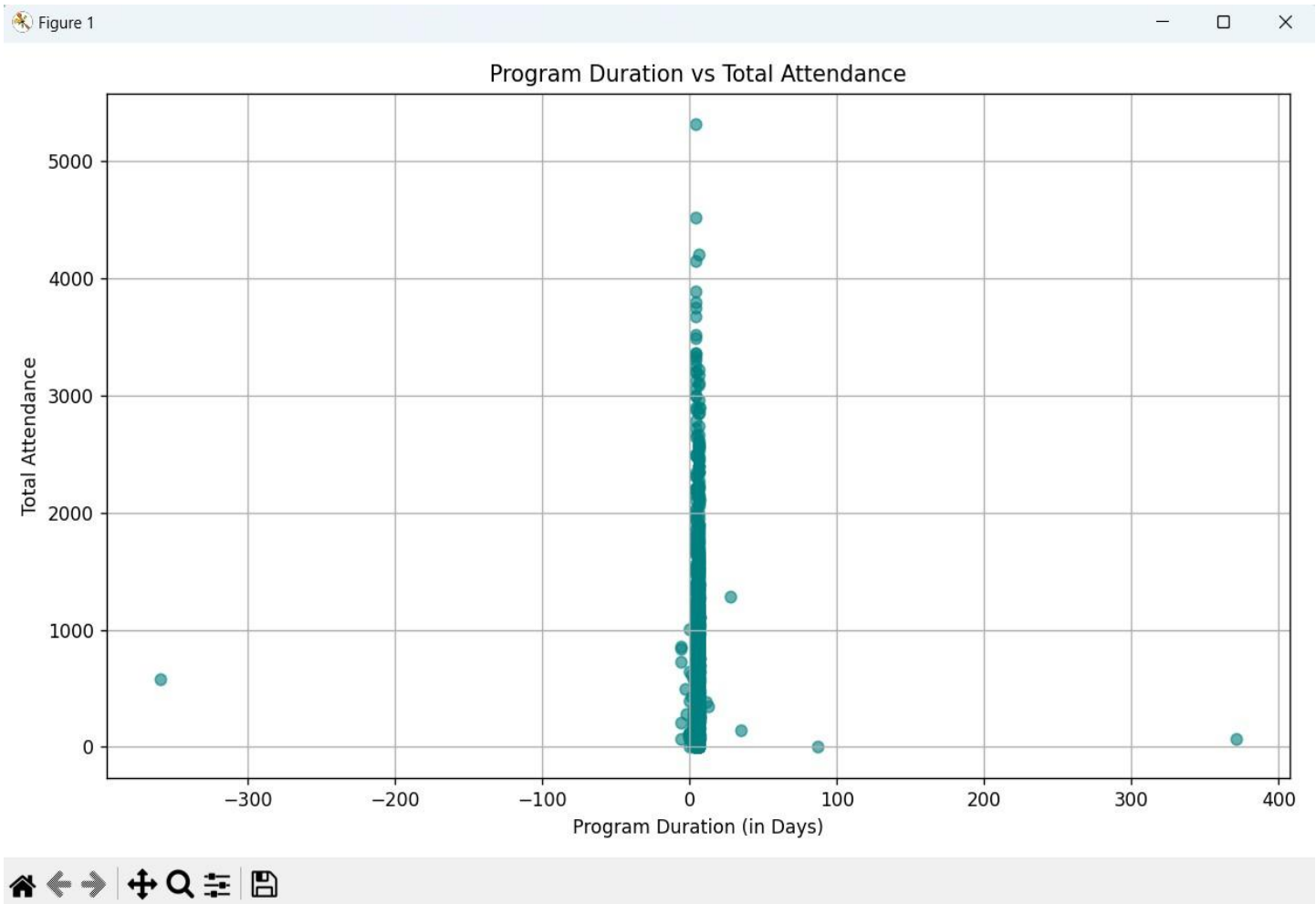
- Used plt.plot() with line color 'red' and markers 'o'
- X-axis labeled as **Year**
- Y-axis labeled as **Number of Programs**

- Grid enabled for readability

4.4.4 Analysis Results

Program offerings saw a **dramatic rise from 2016 to 2017**, reaching a peak. From 2017 onward, there's a **gradual decline**, with the sharpest drop seen between 2020 and 2021. This likely reflects external constraints such as resource limitations or the impact of the COVID-19 pandemic.

4.4.5 Visualization



FUTURE SCOPE

Based on the analysis of the *Kids In Motion: Playground Programming (2016–2021)* dataset, several opportunities exist for expanding insights and improving decision-making through further exploration and data enhancements:

1. Program Scheduling Optimization

With attendance data clearly showing higher engagement on certain days (notably Fridays and Wednesdays), future programs can be strategically scheduled to maximize participation. Conversely, targeted promotions or alternative activities can be tested on traditionally low-attendance days (e.g., Sundays, Mondays) to increase outreach.

2. Borough and Playground-Level Customization

Further granularity in analysis can help identify boroughs and specific playgrounds that either outperform or underperform. This can guide localized program development, staffing, and resource distribution, ensuring community-specific needs are met effectively.

3. Trend Forecasting and Time Series Analysis

Expanding the time frame of the dataset beyond 2021 would enable time series modeling (e.g., ARIMA, Prophet) to forecast future attendance patterns and assess long-term trends, seasonality, and anomalies.

4. Demographic and Sentiment Integration

Incorporating demographic data or participant feedback would allow for clustering analysis and sentiment classification. This could inform content customization, accessibility improvements, and inclusive program design.

5. Resource Allocation and Impact Assessment

Attendance metrics can be tied to budgeting frameworks to evaluate the return on investment (ROI) for different programs. This would help prioritize funding for high-impact activities and explore alternative models in areas with low engagement.

6. External Factor Correlation

Analyzing external variables like weather conditions, public holidays, and school schedules could uncover correlations affecting attendance. Merging with external datasets can enhance model accuracy and support scenario planning.

7. Geospatial Visualization

Using libraries like geopandas, folium, or plotly, future work could map attendance density and identify geographical trends. This spatial analysis would assist in strategic program expansion and resource planning.

8. Interactive Dashboards

Deploying interactive visualizations (e.g., using Dash, Streamlit, or Tableau) could provide stakeholders with realtime access to attendance trends, improving transparency and enabling data-driven decision-making.

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